RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Wednesday, 30 September 2026, 9:32 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:33 PM
Time taken	24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }
    bool *seen = (bool *)calloc(n + 1, sizeof(bool));
    if (seen == NULL) {
        return 1;
    }

    int current_element;
    int duplicate = -1;
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &current_element) == 1) {
            if (seen[current_element]) {
                duplicate = current_element;
            }
            seen[current_element] = true;
        }
    }
    if (duplicate != -1) {
        printf("%d\n", duplicate);
    }
    free(seen);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔

	Input	Expected	Got	
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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